

A General Linear Instability Dispersion Relation for Complex, Multi-Physics Hydrodynamic Flows

Michael D. Walker* & Michael E. Mueller
Princeton University, USA

Abstract

The Rayleigh-Taylor instability is a fundamental hydrodynamic instability arising at the interface between fluids of different densities in an accelerating field. While standard linear theory accurately describes the instability in inviscid, incompressible, and inert fluids, modern applications in inertial confinement fusion (ICF), laboratory astrophysics, and chemically reacting flows require a rigorous treatment of multi-physics effects. This paper presents the derivation and implementation of GLIDE (General Linear Instability Dispersion Equation), a comprehensive 6×6 characteristic determinant that simultaneously accounts for viscosity, rarefaction, surface tension, mass diffusion, compressibility, magnetohydrodynamic tension, heat release from chemical/nuclear reactions, and ablative stabilization. Furthermore, this model is extended to time-varying gravitational acceleration, unifying the Rayleigh-Taylor and Richtmyer-Meshkov regimes via a second-order differential equation. The model reveals the complex interplay between stabilizing forces (viscosity, magnetic tension, ablation) and destabilizing mechanisms (slip, exothermicity), providing a unified model for capturing the dynamic history of ICF implosions from ablative stabilization to reaction-driven stagnation.

1 Introduction

Hydrodynamic instabilities such as Rayleigh-Taylor (RTI) and Richtmyer-Meshkov (RMI) govern the mixing efficiency in high-energy density environments, often causing undesirable radiative losses (Casey (2014)). In the classical RTI limit (inviscid, incompressible, and immiscible), an initial perturbation η_0 grows exponentially $\eta(t) = \eta_0 \exp(\gamma t)$ with the growth rate γ of perturbation wavenumber k as $\gamma = \sqrt{A_t g k}$, where A_t is the Atwood number and g is the acceleration ($A_t g \approx \text{constant}$). In the non-linear regime, the Rayleigh-Taylor instability asymptotically approaches quadratic growth in time $\eta \propto g t^2$, as derived by Ristorcelli and Clark (2004); Cook et al. (2004).

For practical cases of RTI where the initial perturbation is multi-modal (e.g., $\eta_0(x, y) \sim \sum_k \sin(kx) \sin(ky)$ for some range of wavenumbers specified $k \in [k_{\min}, k_{\max}]$), understanding what wavenumber k^* results in the maximum instability growth rate γ^* is critical for initializing simulations and experiments with consistent, replicable initial conditions. In the classical case ($\gamma = \sqrt{A_t g k}$), this is trivial, as it is the largest perturbation wavenumber imposed $k^* = k_{\max}$. However, shown in Table 1, physical effects modify the growth rates and stabilize (restoring and damping forces) or destabilize the instability, particularly at high wavenumbers. Therefore in multi-physics parametric studies, k^* will change based on the range of initial conditions. The ability to calculate k^* from theory is critical to produce replicable multi-modal results, ensuring that the range of wavenumbers specified is less than k^* for the range of initial conditions, so that $k_{\max}$ is truly controlling and representative of maximal growth. This concept is demonstrated visually in Fig. 1, where the range of wavenumbers is sampled from some initial energy distribution $E(k)$ and the scales grow in the transition to turbulence.

*Corresponding author: michael.walker@princeton.edu

Table 1: Summary of literature isolating physical mechanisms affecting RTI growth rate.

Physical Mechanism	Growth Modification ($\Delta\gamma$)	Role	Dominant Regime	Key References
Classical Inviscid	$\gamma = \sqrt{A_i g k}$	Baseline	All k	Strutt (1883); Taylor (1950)
Viscosity	$\gamma \sim -\mu k^2/\rho$	Stabilizing	High k	Bellman and Pennington (1954); Chandrasekhar (1981)
Surface Tension	$\gamma \sim -\sqrt{\sigma k^3/\rho}$	Stabilizing	High k (Cutoff)	Chandrasekhar (1981); Sohn (2009)
Mass Diffusion	$\gamma \sim -\mathcal{D}k^2$	Stabilizing	High k	Duff et al. (1962)
Ablation	$\gamma \sim -kV_a$ (Convection)	Stabilizing	High k	Takabe et al. (1985); Clavin and Masse (2004); Huntington (2018)
Density Gradient	$\gamma \sim \sqrt{k/(1+kL_\rho)}$, $L_\rho \equiv \rho/ \nabla\rho $	Stabilizing	High k	LeLevier et al. (1955); Munro (1988); Mikaelian (2005)
Magnetic Tension	$\gamma \sim -(\vec{k} \cdot \vec{B})^2/\rho$	Variable ¹	High k (Anisotropic)	Carlyle and Hillier (2017); Chandrasekhar (1981)
Compressibility	$A_{\text{eff}} < A_i$ (Reduced Drive)	Stabilizing	Low k	Bernstein and Book (1983); Fu et al. (2025)
Rarefaction (Slip)	$\gamma \sim +kL_s$ (Reduced Drag)	Destabilizing	High k	Majumder et al. (2024)
Reactive Heat Release	Exothermic Expansion	Destabilizing	Reaction Width	Sanz et al. (2002); Clavin et al. (2005); Bychkov et al. (2015)
Time-Varying $g(t)$	Dynamic History	Variable ²	All k	Schilling (2024)

1. Based on orientation of the magnetic field lines with respect to the interface.
2. Based on whether the heavy fluid is accelerating or decelerating into the light fluid.

Extreme physical regimes like Inertial Confinement Fusion (ICF) involve a trajectory through parameter space where the classical approximation fails. An ICF implosion proceeds in phases, each dominated by different physical mechanisms:

1. **Acceleration phase:** The shell is accelerated inward by *ablation*. The mass *diffusion* carries vorticity away from the interface (“rocket effect”), providing a critical stabilizing term.
2. **Coasting phase:** As the laser turns off, the shell coasts. Instabilities may continue to grow due to *time-varying acceleration*, though magnetic fields and interfacial surfaces can provide *tension*.
3. **Stagnation phase:** Near peak compression, densities are high and the fluid is highly *compressible*. If ignition occurs, *nuclear heat release* creates a reactive pressure that can violently destabilize the interface.

Existing analytic models typically treat these physical effects in isolation. Table 1 shows how physical effects modify the growth rates and stabilize (restoring and damping forces) or destabilize the instability, particularly at high wavenumbers. This work synthesizes these disparate models into a single, generalized dispersion relation, GLIDE (General Linear Instability Dispersion Equation), allowing for the continuous integration of the instability amplitude $\eta(t)$ as the physical parameters evolve.

2 Physical model and linearization

This analysis considers two semi-infinite fluid layers separated by an equilibrium interface $z = 0$. The fluids are characterized by densities $\rho_{1,2}$, viscosities $\mu_{1,2}$, mass diffusion coefficients $\mathcal{D}_{1,2}$, sound speeds $a_{1,2}$, and magnetic permeability μ_m . The system is subject to a vertical acceleration $\vec{g}(t)$, uniform magnetic field $\vec{B}$, chemical heat release $\dot{Q}_{\text{chem}}$, and, at the interface, a surface tension σ and an ablation velocity V_a (with performance coefficient $\beta \approx 3\text{--}4$). The dynamics are governed by the compressible Navier-Stokes equations coupled with ideal magnetohydrodynamics and Fickian diffusion.

Normal mode perturbations of the form $\eta(x, z, t) = \eta_0(z) + \hat{\eta}(z) \exp(ikx + \gamma t)$ are applied, where γ is the complex growth rate. Substituting these into the governing equations and neglecting second-order terms yields three distinct decay mechanisms in the bulk fluid:

1. **Compressible-Potential Mode (m):** $m = \sqrt{k^2 + \gamma^2/a^2}$, capturing acoustic response. Assuming irrotational flow and substituting the continuity equation into the momentum equation yields the pressure wave equation. In the incompressible limit ($a \rightarrow \infty$), $m \rightarrow k$. Solutions are of the form $\exp(\pm mz)$.
2. **Diffusive Mode (n):** $n = \sqrt{k^2 + \gamma/\mathcal{D}}$, capturing species transport. From the mass continuity and species transport equations, solutions are of the form $\exp(\pm nz)$.
3. **Viscous-Vorticity Mode (q):** $q = \sqrt{k^2 + \rho\gamma/\mu}$, capturing rotational shear. From curl of the momentum equation, solutions are of the form $\exp(\pm qz)$.

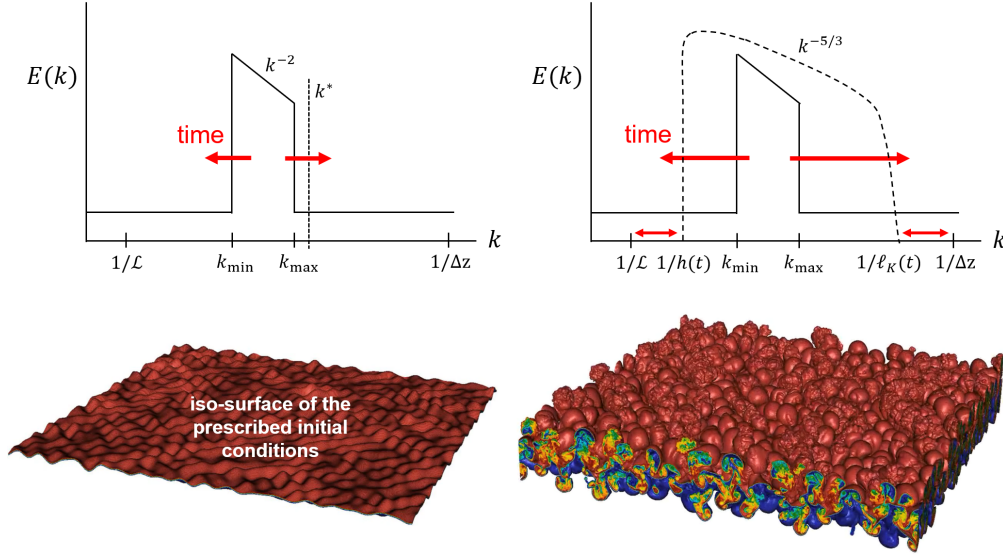


Figure 1: Here a chosen initial distribution $E(k) \propto k^{-2}$ favors smaller wavenumbers initially. A transition to Kolmogorov $k^{-5/3}$ turbulence in time is assumed, as the mixing layer height h approaches the domain size $\mathcal{L}$ and the Kolmogorov scale ℓ_K approaches the grid resolution Δz .

The general solution may then be constructed such that the solution for the appropriate fluid vanishes at $\pm\infty$:

$$\hat{w}_1(z) = A_1 e^{m_1 z} + B_1 e^{q_1 z} + C_1 e^{n_1 z}, \quad \hat{w}_2(z) = A_2 e^{-m_2 z} + B_2 e^{-q_2 z} + C_2 e^{-n_2 z},$$

where $\hat{w}$ is the vertical velocity eigenfunction, and the horizontal component $\hat{u}$ is obtained through the continuity relation $ik\hat{u} = -d\hat{w}/dz$ (in the incompressible approximation for the viscous mode).

3 General dispersion matrix and boundary conditions

Stability is determined by satisfying conservation laws across the interface, resulting in a system of linear equations $\mathbb{M}\mathbf{x} = 0$, where the vector $\mathbf{x} = [A_1, B_1, C_1, A_2, B_2, C_2]^T$ corresponds to the amplitudes of the potential, viscous, and diffusive modes for both fluids. The 6×6 matrix $\mathbb{M}$ incorporates several high-order boundary conditions. The first row is kinematic continuity across the interface with ablative mass flux $\rho_1(\hat{w}_1 - \gamma\eta) = \rho_2(\hat{w}_2 - \gamma\eta)$. Row 2 is the rarefied slip condition. For rarefied flows (slip Knudsen number $\text{Kn}_s \equiv kL_s > 0.01$), a generalized Navier slip condition is applied: $u_2 - u_1 = L_s \tau_{xz}$, where L_s is the slip length and $\tau_{xz} = \mu(du/dz + ikw)$ is the shear stress. This introduces a factor $\Phi = L_s \mu_2(q_2^2 + k^2) \approx kL_s(\mu_2/\mu_1)$, capturing the jump in horizontal velocity due to rarefaction, reducing the shear stress transfer across the interface. This “lubrication” effect is destabilizing at high wavenumbers. Row 3 is the shear stress continuity $\mu_1(d^2 w_1/dz^2 + k^2 w_1) = \mu_2(d^2 w_2/dz^2 + k^2 w_2)$. Rows 5 and 6 enforce the continuity of species concentration $\hat{Y}_1 = \hat{Y}_2$ and mass flux $\rho_1 \mathcal{D}_1 \nabla \hat{Y}_1 = \rho_2 \mathcal{D}_2 \nabla \hat{Y}_2$. Because the concentration perturbation is driven by the vertical convection of the base state gradient, $\hat{Y} \propto \hat{w}/\gamma$.

Row 4 is the normal stress balance, where the bulk stresses from the fluids are balanced by the forces of interface tension. The generalized tension $\mathcal{T}_{\text{tot}}$ unifies all forces that act on the interface:

$$\mathcal{T}_{\text{tot}} = \underbrace{\frac{\sigma k^2}{\gamma}}_{\text{Surface}} + \underbrace{\frac{(\vec{k} \cdot \vec{B})^2}{\mu_m k \gamma}}_{\text{Magnetic}} + \underbrace{\frac{\beta k \dot{m} V_a}{\gamma}}_{\text{Ablation}} + \underbrace{\frac{\dot{Q}_{\text{chem}}}{\gamma}}_{\text{Reaction}}. \quad (1)$$

The stress operator $\mathcal{S}$ represents the total normal stress contribution from a specific mode (potential or

viscous, $\lambda = m, q$) within each fluid layer ($j = 1, 2$):

$$\mathcal{S}_j(\lambda) = \underbrace{-\frac{\rho_j \gamma}{k}}_{\text{Dynamic}} + \underbrace{\frac{\rho_j g}{\gamma}}_{\text{Hydrostatic}} - \underbrace{2\mu_j \lambda k}_{\text{Viscous}}. \quad (2)$$

The full dispersion relation is obtained by setting $\det(\mathbb{M}) = 0$, where

$$\mathbb{M} = \begin{bmatrix} \rho_1 & \rho_1 & \rho_1 & -\rho_2 & -\rho_2 & -\rho_2 \\ m_1/k & k/q_1 & 0 & m_2/k + \Phi & k/q_2 + \Phi & 0 \\ 2\mu_1 m_1 k & \mu_1(k^2 + q_1^2) & 0 & -2\mu_2 m_2 k & -\mu_2(k^2 + q_2^2) & 0 \\ \mathcal{S}_1(m_1) & \mathcal{S}_1(q_1) & \mathcal{T}_{\text{tot}} & \mathcal{S}_2(m_2) & \mathcal{S}_2(q_2) & -\mathcal{T}_{\text{tot}} \\ \gamma^{-1} & \gamma^{-1} & \gamma^{-1} & -\gamma^{-1} & -\gamma^{-1} & -\gamma^{-1} \\ 0 & 0 & \rho_1 \mathcal{D}_1 n_1 & 0 & 0 & -\rho_2 \mathcal{D}_2 n_2 \end{bmatrix}. \quad (3)$$

4 Dynamic implementation

For ICF applications, the static growth rate derived from $\det(\mathbb{M}) = 0$ is insufficient because parameters like $g(t)$ change on timescales comparable to the instability growth. A second-order oscillator equation is derived by interpreting the dispersion relation as a characteristic polynomial for the differential operator d/dt with the WKB-like approximation $\gamma^2 \leftrightarrow (d^2\eta/dt^2)/\eta$:

$$\frac{d^2\eta}{dt^2} + 2\Omega_{\text{eff}} \frac{d\eta}{dt} - \mathcal{F}(k, t) \eta(t) = 0. \quad (4)$$

Using impulsive acceleration $g(t) = u_{ps} \delta(t)$ and integrating Eq. 4: $[d\eta/dt]_{0^-}^{0^+} \approx \mathcal{F}(k, t) = k A_{t,ps} u_{ps} \eta_0 + \dots$ recovers the classical Richtmyer-Meshkov linear growth rate (Thornber (2017)) to leading order. Here, $A_{t,ps}$ is the post-shock Atwood number, u_{ps} is the induced velocity, and $\delta(t)$ is the Dirac delta function.

The effective damping term Ω_{eff} sums the viscous, diffusive, and ablative losses. Crucially, slip reduces the effective viscosity:

$$2\Omega_{\text{eff}} = \frac{2k^2(\mu/\rho + \mathcal{D})}{1 + kL_s} + 2\beta k V_a. \quad (5)$$

Ablation (kV_a) acts as a linear-in- k damping, while viscosity is quadratic. So for intermediate wavelengths, ablative damping is often the dominant stabilizing mechanism.

The effective drive term $\mathcal{F}$ accounts for the reduction in buoyancy due to compressibility:

$$\mathcal{F}(k, t) = A_{\text{eff}}(t)g(t)k - \frac{(\vec{k} \cdot \vec{B})^2}{\mu_m(\rho_1 + \rho_2)} - \frac{\sigma k^3}{\rho_1 + \rho_2} + \frac{\dot{Q}_{\text{chem}}}{\rho_1 + \rho_2}, \quad (6)$$

where $A_{\text{eff}}(t) = A_t/(1 + g(t)/ka^2)$ is an effective Atwood number (Bernstein and Book (1983)). As the fluid becomes compressible ($a \rightarrow 0$), the effective Atwood number drops, as energy is expended on compression rather than interface displacement.

5 Computational implementation

The GLIDE framework is implemented in Python¹ using a dual-numerical strategy.

- **Static roots:** A Newton method with secant approximation finds the complex roots γ of $\det(\mathbb{M}) = 0$. The solver is seeded with the inviscid growth rate $\gamma_0 = \sqrt{A_t g k}$ to avoid convergence to trivial roots.
- **Dynamic integration:** For time-varying gravity, Eq. 4 must be integrated and may become stiff, particularly at high wavenumbers where restoring forces and damping terms grow rapidly. An explicit Runge-Kutta method (RK45) is employed. Adaptive step-size is crucial for capturing the rapid transient growth during shock impulses while efficiently stepping through the coasting phases.

¹A numerical solver is in development at the following repository: <https://github.com/mw6136/GLIDE>.

6 Conclusion

This work derives a comprehensive dispersion relation that considers viscosity, rarefaction, surface tension, compressibility, diffusion, magnetohydrodynamics, ablation, and heat release, unifying disparate physical effects into a single matrix formulation. The resulting model, GLIDE, allows for the prediction of instability growth in multi-physics regimes where continuum or inviscid approximations fail. The dynamic formulation further bridges the gap between constant-acceleration Rayleigh-Taylor instability and impulsive Richtmyer-Meshkov instability.

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